



Velocity Based Training

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What is VBT?

- Technology that measures the speed at which a person is training at.
- Positive response to training (Zhang et al, 2022).



Literature

Positive effects on competition season training stimulus, reduced training stress and power specific adaptations (Orange et al, 2019).

Male beach volleyball players saw an increase in CMJ with a 10% velocity loss cut off for quarter squats (Aquino et al, 2025).

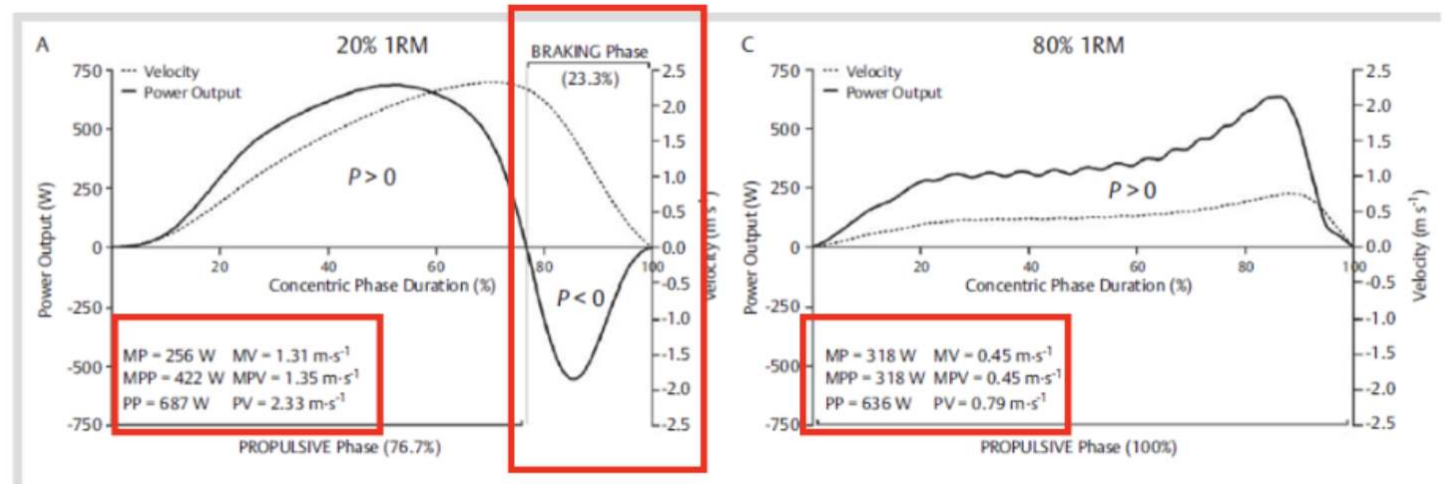
Percentage based training and VBT training yielded the same strength results, but VBT induced greater gains in peak velocity of CMJ, sprinting speed, and change of direction (Banyard et al, 2021).

Velocity loss as little as 10% showed increases in neuromuscular gains (Rodríguez-Rosell et al, 2020) with no significant difference in strength gains (Pareja-Blanco et al, 2020).

Mean Propulsive Velocity (MPV) vs Mean Velocity (MV)

Figure 4. Concentric phase of one repetition with 20% 1RM and one with 80%1RM in the bench press exercise

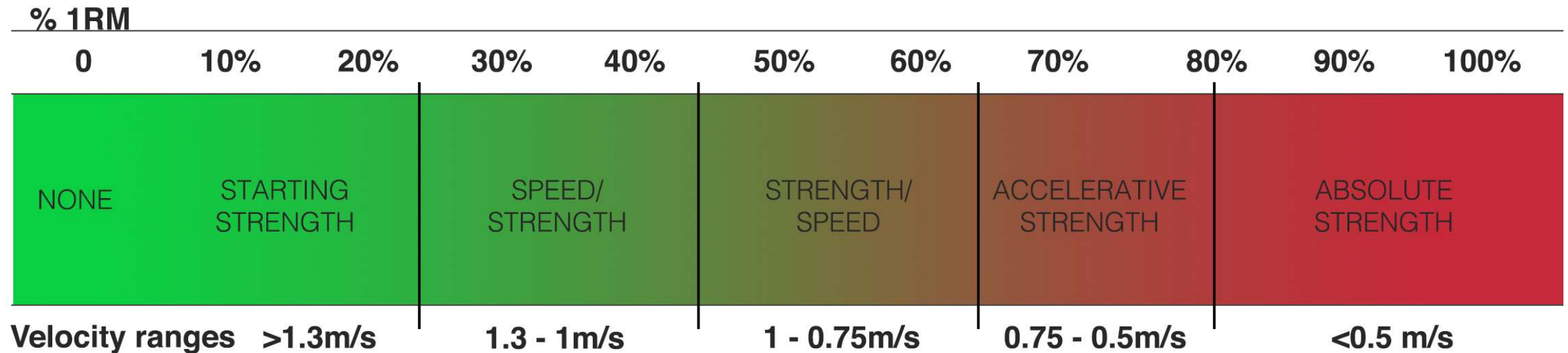
- MPV → lighter loads and accurate relative load to 1RM (González-Badillo & Sánchez-Medina, 2010)
- MV → ballistic and heavy loads (García-Ramos et al, 2018).



Adapted from Sanchez-Medina et al. (2010)

Velocity Zones

VELOCITY ZONES



Load Velocity Profile

**MEAN PROPULSIVE VELOCITY (m/s)
FOR EACH % OF 1RM**

VBT

VITRUVUE
VBT FOR YOU

		1RM	95%	90%	85%	80%	75%	70%	65%	60%	55%	50%	45%	40%
Bench Pull		Sánchez Medina et al (2014) Prone lying rowing on bench with Smith machine												
		0.53	0.59	0.65	0.72	0.78	0.85	0.92	0.99	1.06	1.13	1.21	1.28	1.36
Back Squat		Sánchez Medina et al (2014) Deep squat with Smith Machine and pauses												
		0.32	0.42	0.51	0.59	0.68	0.76	0.84	0.92	1.00	1.07	1.14	1.21	1.28
Pull Up		Sánchez Moreno et al. (2017) Prone Pull up												
		0.22	0.31	0.39	0.50	0.57	0.65	0.74	0.83	0.91	1.00	1.09	-	-
Military Press		Muñoz et al. (2014) Militar Press behind the neck with Smith Machine												
		0.20	0.27	0.34	0.41	0.49	0.55	0.62	0.69	0.75	0.81	0.86	0.92	0.97
Bench Press		González Badillo and Sánchez Medina (2010): Bench press with Smith machine and pauses												
		0.18	0.25	0.32	0.39	0.47	0.55	0.62	0.70	0.78	0.87	0.95	1.07	1.13
Deadlift		Helms et al. 2017: Dead-lift with expert powerlifters 1RM=237,3kg												
		0.14	0.21	0.29	0.37	0.46	-	-	-	-	-	-	-	-
Hip Thrust		Hoyo, M, Núñez et al (2017) Predicting Loading Intensity Measuring Velocity in Barbell Hip Thrust Exercise												
		0.24	-	0.36	-	0.48	-	0.6	-	0.72	-	-	-	-

Velocity Loss (VL)

10% VL and 30% VL = same hormonal response, 10% VL = greater neuromuscular gains, no significant strength gains differences
(Rodríguez-Rosell et al, 2020, Pareja-Blanco et al, 2020)

Group	70% of 1RM AVG Reps = 9.6			75% of 1RM AVG Reps = 7.7			80% of 1RM AVG Reps = 6.0			85% of 1RM AVG Reps = 4.5		
VL	Reps	RIR	RPE	Reps	RIR	RPE	Reps	RIR	RPE	Reps	RIR	RPE
0	1.0 ± 0.0	8.6	1.4	1.0 ± 0.0	6.7	3.3	1.0 ± 0.0	5.0	5	1.0 ± 0.0	3.5	6.5
10	3.7 ± 0.9	5.9	4.1	3.1 ± 0.8	4.6	5.4	2.5 ± 0.9	3.5	6.5	2.2 ± 0.9	2.3	7.7
20	4.2 ± 1.2	5.4	4.6	3.8 ± 1.2	3.9	6.1	2.9 ± 0.8	3.1	6.9	2.3 ± 0.8	2.2	7.8
40	7.6 ± 2.5	2.0	8.0	7.0 ± 2.3	0.7	9.3	5.3 ± 1.5	0.7	9.3	4.3 ± 1.5	0.2	9.8

Data from Pareja-Blanco et al. (2020); cross-referenced with data from Rodríguez-Rosell et al. (2019).
VL = Velocity Loss; RIR = Repetitions in Reserve; RPE = Rating of Perceived Exertion; AVG = Average.

What does it all mean?

- Minimum effective dosage
- Individual feedback to daily training load and intensity
- Limiting overtraining and non-functional overreaching

Strengths and Limitations

Strengths

- Safe and effective
- Athlete conversations
- Individual
- Instant feedback

Limitations

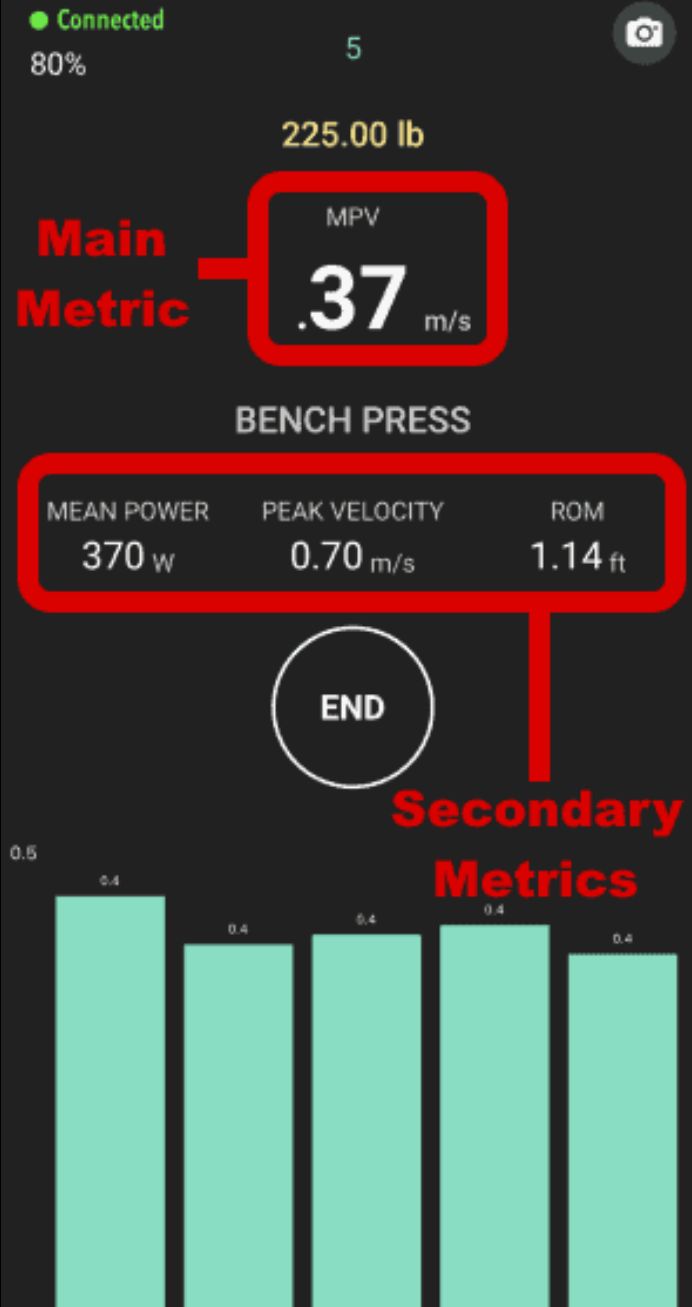
- Time
- Cost

Future Research

- Female data



Part 2: Future Application



Colby-Sawyer
VBT





Hex Bar Deadlift

4 x 5, 5, 4, 2 - Stay within 10 lbs of the percentages, talk to a coach if you have questions



☐ Update all future occurrences of this exercise on this calendar

Exercise Name

Hex Bar Deadlift

Sets

4

Reps/Time

Reps

CUST

☐ Each Side

Left & Right Side

Custom Reps

5

Set 1

5

Set 2

4

Set 3

2

Set 4

☒ Custom Percentages

%

Set 1

%

Set 2

%

Set 3

%

Set 4

☐ Prescribe Load

Additional Information (optional)

Stay within 10 lbs of the percentages, talk to a coach if you have

Instructions specific to this workout (max 255 chars.)

Workout Grouping

B

Used for Supersets

Tempo

Exercise tempo

Rest Time

120

Rest (secs) between sets

Additional Options

☐ Completion Lift

No user input

☐ Body Weight Lift

Input # of Reps only

☐ Coach Completion

Coach inputs data

☐ Disable Max Tracking

Will not update max

☐ Force Max/PR Update

Replaces current Max/PR

☐ Peak Power

☒ Bar Speed

Adds m/s input

☒ Track Rep Count

☒ Track Volume Load



SUPERSET 1 OF 2

REPS

PEAK
VELOCITY

200

5

0.81

(L) REST TIMER (120 SECS)

210

5

0.72

(L) REST TIMER (120 SECS)

220

4

0.56

(L) REST TIMER (120 SECS)

230

2

0.45

In-Season Training, Power Adaptation

History

Hex Bar Deadlift

1	2	3	4
200	210	220	230
5	5	5	4
0.83	0.79	0.65	0.54

ormation

17
DEC
WED
2025

1	2	3	4
200	210	220	230
5	5	4	2
0.81	0.72	0.56	0.45

formation

17
DEC
WED

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